BCS-503	JAVA PROGRAMMING	L T P 1 1 0	2 credits

Pre-requisites: Computer Fundamentals & Principle of Computer Programming, Programming Concepts of C and C++

Course Objectives:

CO1	To understand object-oriented concepts
CO2	To learn the basic concept of JAVA language.
CO3	To learn how to design GUI applications.
CO4	To understand the concept of JDBC
CO5	To learn to build applications in JAVA language

Detailed Syllabus

MODULE-I

The Java Language: History and evolution of Java, Java's Lineage, The Creation of Java, Java's Magic Code; The Byte Code, The Java's Class File Format, The java's Buzzwords, The Evolution of Java. Object Orientation concepts; Class, Object and its significance. Environment variable. Data Types, Variables and Array: Strongly typed Language, Primitive type, Non Primitive type, Wrapper classes, Scope & lifetime of the variables, Type Conversion and casting, Automatic Type promotions, Operators: Arithmetic operator, The Bitwise operator, Relational operator, Assignment operator, The ?Operator, Operator precedence. Control Statements: Selection Statement, Iteration Statement, Jump Statement. **Introducing classes:** Class Fundamentals, Object Garbage Collection, Creating and Operating Objects, methods, Nested, Inner Class &Anonymous Classes, Abstract Class & Interfaces, Defining Methods, Argument Passing Mechanism, Method Overloading, Recursion, Garbage collection, Finalize () Method.

Module II

Inheritance: Use and Benefits of Inheritance in OOP, Types of Inheritance in Java, Inheriting Data Members and Methods. Role of Constructors. Overloading concept &. Polymorphism in inheritance. **Package:** Organizing Classes and Interfaces in Packages. Package as Access Protection Defining Package CLASSPATH Setting for Packages, **Exception Handling:** The Idea behind Exception ,Exceptions & Errors Types of Exception, Control Flow In Exceptions, JVM reaction to Exceptions, Use of try, catch, finally, throw, throws in Exception Handling, Inbuilt and User Defined Exceptions, Checked and Un-Checked Exceptions, **Thread:** Understanding Threads, Needs of Multi-Threaded Programming, Thread Life-Cycle, Thread Priorities, Synchronizing Threads, Inter Communication of Threads, **Array &String :**Defining an Array, Initializing & Accessing Array, Multi–Dimensional Array, Operation on String

MODULE-III

Applet: Applet & Application, Applet Architecture, Parameters to Applet,. **Event Handling:** Event-Driven Programming in Java, Event Handling Process, **GUI Programming (Java AWT):**Components and Containers: Basics of Components, Using Containers, Layout Managers, AWTComponents, Adding a Menu to Window Using Swing Components., Introducing Swings: The Origin of swings, swings is built on AWT, Swings package & event Handling. Database Programming using JDBC: Introduction to JDBC, JDBC Drivers

Text Book:

1. Herbert Schieldt, "The Complete Reference: Java" Seventh Edition, TMH. Reference Books:

1. Herbert Schieldt" Java Programming Cook Book" McGraw Hill.
2. Core Java™ 2 Volume I - Fundamentals, Seventh Edition Prentice Hall PTR
3. Core Java™ 2 Volume II - Fundamentals, Seventh Edition Prentice Hall PTR

CO1	Design the process of interaction between Objects and System w.r.t. Object Oriented Paradigm.
CO2	Acquire a basic knowledge of Object Orientation with different properties as well as different features of Java, threads
CO3	Analyze basic programming concepts in Java with different object related issues and various string handling functions as well as basic I/O operations
CO4	Discuss basic Code Reusability concept w.r.t. Inheritance, Package and Interface
CO5	Implement Exception handling, Multithreading and Applet (Web program in java) programming concept in Java